# Technical design document

# Project: AXM Vehicle Manufacturer

Document history

| Version | Author | Comment |
|---|---|---|
| 26.05.2022 | Stefan Ristovski | Version 1.0 |
| | | |

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## Project description

In this project we will explore the process of manufacturing one vehicle. To achieve the required process, we will implement solution through Dynamics 365 Finance & Operations. The module that we will create will be called AXM Vehicle Manufacturer.

While working on this project we will get familiar with the capabilities of the Dynamics 365 Finance & Operations as how to create a different type of Forms, Tables, EDTs, Enums, Reports, Menus, Menu Items and Security as well. But the main thing we will experiment with is extensions. This is unique feature compared to the previous versions of AX.

## Base Enums

| Name | Label | Values |
|---|---|---|
| AXMVehiclePart | Vehicle part | Door, Engine, Wheels, Chassis |
| AXMVehicleType | Vehicle type | Sedan, Coupe, SUV, Hatchback, Truck |
| AXMStatus | Status | Open, Validated, Completed |

## Extended Data Types

| Name | Extends | Type | Label |
|---|---|---|---|
| AXMVehiclePartId | | String | Vehicle part identification |
| AXMPartQty | InventQty | Real | Part quantity |
| AXMVehicleManufacurer | VendAccount | String | Vehicle manufacturer |
| AXMVehicleColor | EcoResColorName | String | Vehicle color |
| AXMCOEmissionValue | | Real | CO emission threshold |
| AXMVehicleId | | String | Vehicle identification |

# Table

## Vehicle manufacturer parameters - AXMVehicleManufacturerParameters

Parameters tables are used for configuring setups for a particular module. It always contains only one record, since only one setup can be active per module. In this table we will specify the minimum CO emission value for a car to be build.

### Properties

| Properties | |
|---|---|
| Name | AXMVehicleManufacturerParameters |
| FormRef | AXMVehicleManufacturerParameters |
| Label | Vehicle manufacturer parameters |
| Table Group | Parameters |
| Modified Date Time | Yes |
| Modified By | Yes |
| Created Date Time | Yes |
| Created By | Yes |

### Fields

| Name | Field type | EDT | Properties |
|---|---|---|---|
| COEmissionValue | Real | AXMCOEmissionValue | AllowNegative: No |
| ManufacturingDateLimit | Date | TransDate | HelpText: Vehicle date shouldn't be older than the date specified in the parameters form. |
| Key | Integer | ParametersKey | AllowEdit: No, AllowEditOnCreate: No<br>Visible: No<br>Mandatory: Yes |

## Indexes

| Name | Field list | Properties |
|---|---|---|
| KeyIdx | Key | AllowDuplicates: No<br>AlternateKey: Yes |

Methods to be implemented: **find**().

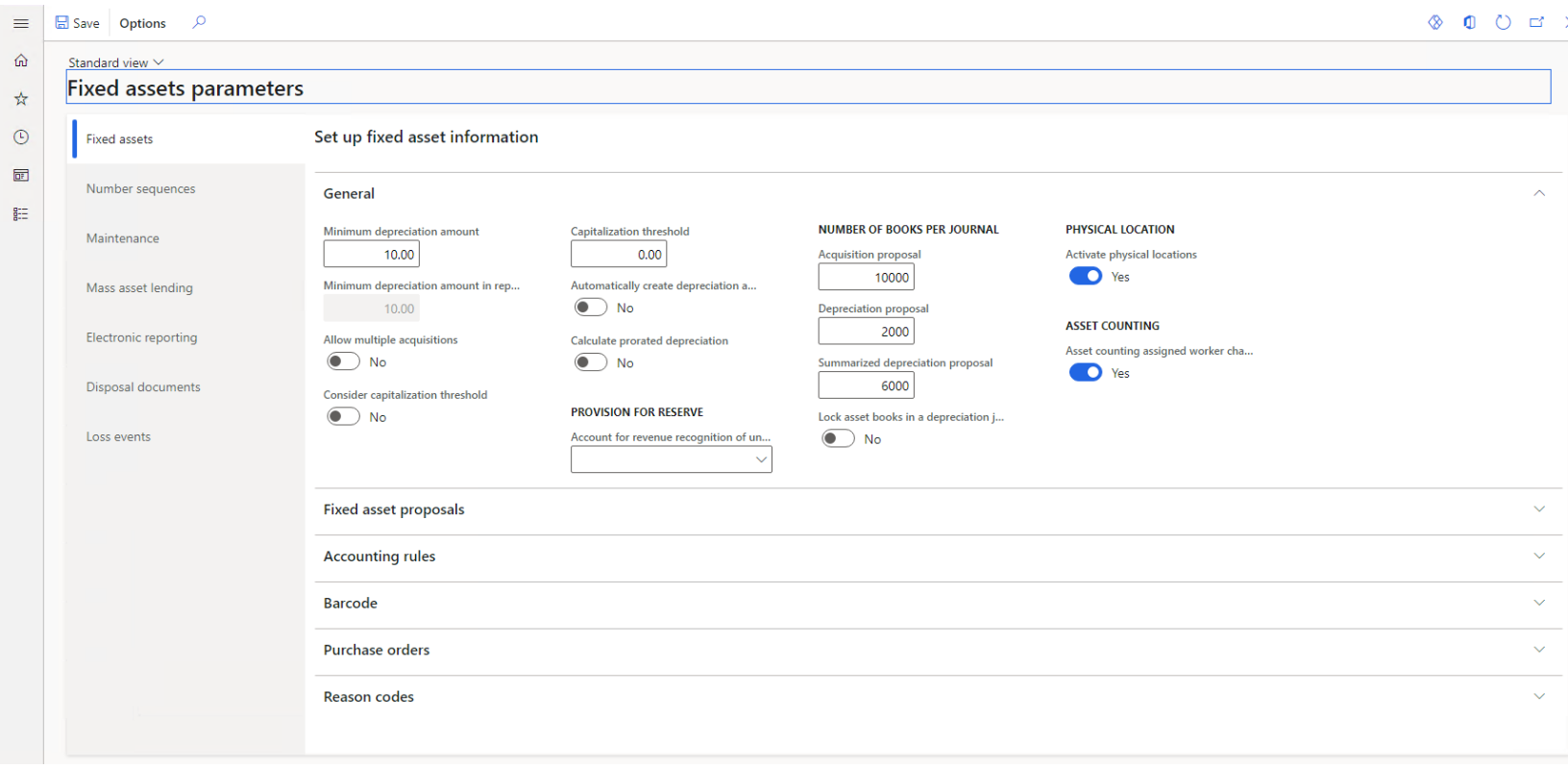
> The find method for parameter table is slightly different than find methods on regular tables. Please check AssetParameter table as example of what should be in the method.

# Form

| Name | Form Template | Label |
|---|---|---|
| AXMVehicleManufacturerParameters | TableOfContents | Vehicle manufacturer parameters |

**DataSource:** AXMVehicleManufacturerParameters

The form should look like: AOT/Forms/AssetParameters




All fields from the table should be in General Fast tab.

# Table

## Vehicle parts - AXMVehicleParts

In this table we will keep information about available vehicle parts from different vendors and their quantities in storage.

### Properties

| Properties | |
|---|---|
| Name | AXMVehicleParts |
| FormRef | AXMVehicleParts |
| Label | Vehicle parts |
| Table Group | Group |
| Modified Date Time | Yes |
| Modified By | Yes |
| Created Date Time | Yes |

| | |
|---|---|
| Created By | Yes |

## Fields

| Name | Field type | EDT | Properties |
|---|---|---|---|
| VehiclePartId | String | AXMVehiclePartId | Mandatory: Yes |
| VehiclePart | Enum | AXMVehiclePart | Mandatory: Yes |
| VehiclePartQty | Real | AXMPartQty | Mandatory: Yes<br>Allow Negative: No |
| VehicleType | Enum | AXMVehicleType | Mandatory: Yes |
| VehiclePartSerialId | String | InventSerialId | Mandatory: Yes |
| VehiclePartManufacurer | String | AXMVehicleManufacurer | Mandatory: Yes<br>Relation:<br>VendTable.AccountNum<br>Label: Vehicle part manufacturer |
| VehiclePartPrice | Real | Price | Label: Vehicle part manufacturer |

## Indexes

| Name | Field list | Properties |
|---|---|---|
| VehiclePartIdx, | VehiclePartId | AllowDuplicates: No<br>AlternateKey: Yes |

**Additional requirements:**

Methods to be implemented: **find**() and **exists**() and display method **displayManufacurerName()** for VehiclePartManufacurer name.

VehiclePartId needs to be number sequence in format: VPID-XXXXX.

**Field Groups:**

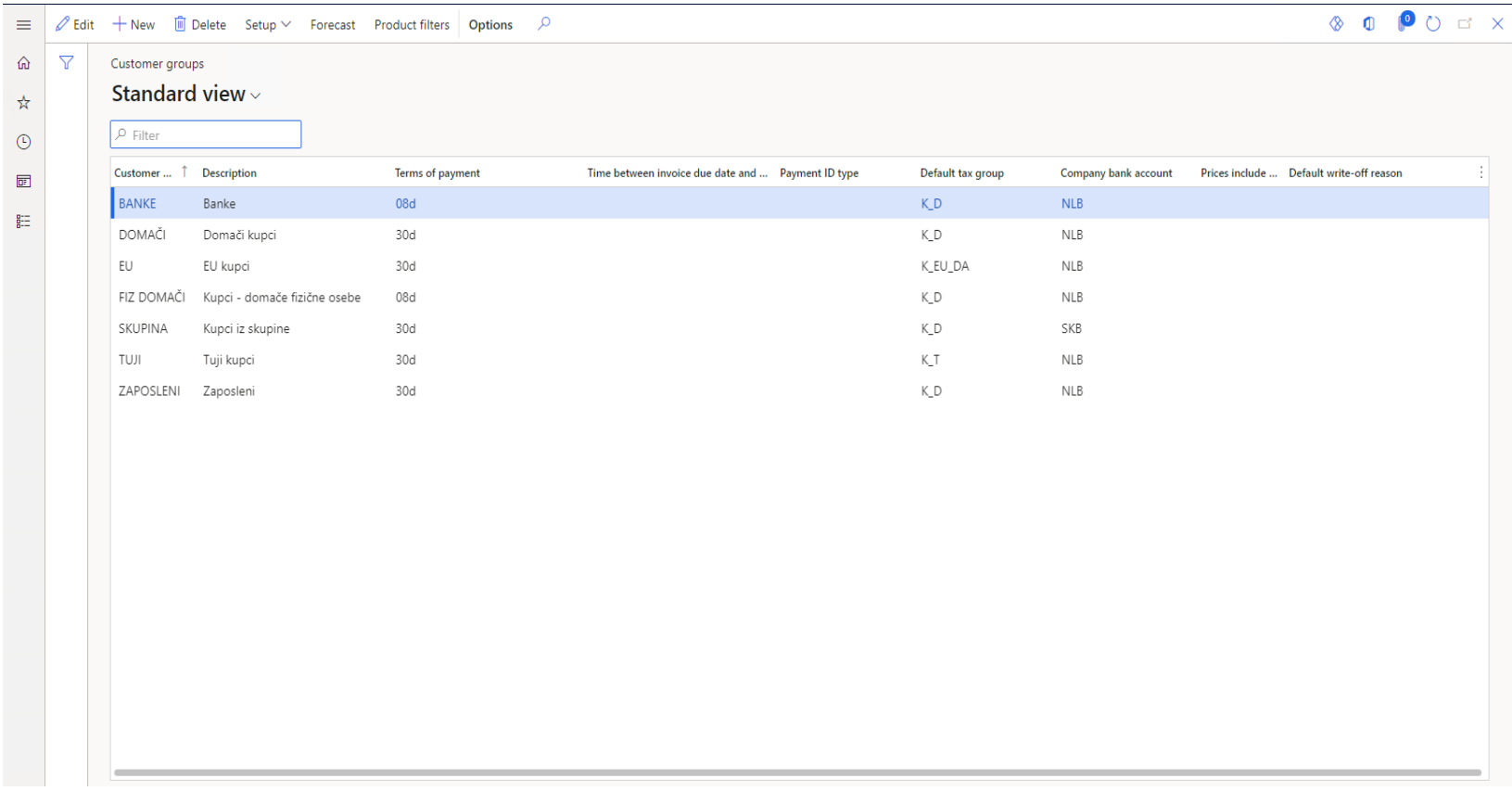
Define them by your own logic of how things should work.

# Form

| Name | Form Template | Label |
|---|---|---|
| AXMVehicleParts | SimpleList | Vehicle parts |

**DataSource:** AXMVehicleParts

The form should look like: AOT/Forms/CustGroup




| Customer ... | Description | Terms of payment | Time between invoice due date and ... | Payment ID type | Default tax group | Company bank account | Prices include ... | Default write-off reason |
|---|---|---|---|---|---|---|---|---|
| BANKE | Banke | 08d | | | K_D | NLB | | |
| DOMAČI | Domači kupci | 30d | | | K_D | NLB | | |
| EU | EU kupci | 30d | | | K_EU_DA | NLB | | |
| FIZ DOMAČI | Kupci - domače fizične osebe | 08d | | | K_D | NLB | | |
| SKUPINA | Kupci iz skupine | 30d | | | K_D | SKB | | |
| TUJI | Tuji kupci | 30d | | | K_T | NLB | | |
| ZAPOSLENI | Zaposleni | 30d | | | K_D | NLB | | |

# Table

## Vehicle type parts – AXMVehicleFormula

In this table we will keep the essential specifications for different vehicle types. This will be used for validating if we have all the required parts for one vehicle to be manufactured. For example, for vehicle type Coupe we need one engine, four wheels etc.

### Properties

| Properties | |
|---|---|
| Name | AXMVehicleFormula |
| FormRef | AXMVehicleFormula |
| Label | Vehicle type parts |
| Table Group | Group |
| Modified Date Time | Yes |

| | |
|---|---|
| Modified By | Yes |
| Created Date Time | Yes |
| Created By | Yes |

### Fields

| Name | Field type | EDT | Properties |
|---|---|---|---|
| VehicleFormulaType | String | AXMVehicleType | Mandatory: Yes |
| VehicleEngine | Enum | NoYesId | Label: Engine |
| VehicleChassis | Enum | NoYesId | Label: Chassis |
| VehicleDoors | Integer | FullQuantity | Allow Negative: No<br>Label: Doors |
| VehicleWheels | Integer | FullQuantity | Allow Negative: No<br>Label: Wheels |

### Indexes

| Name | Field list | Properties |
|---|---|---|
| VehicleFormulaTypeIdx | VehicleFormulaType | AllowDuplicates: No<br>AlternateKey: Yes |

**Additional requirements:**

Methods to be implemented: **find**() and **exists**().

**Field Groups:**

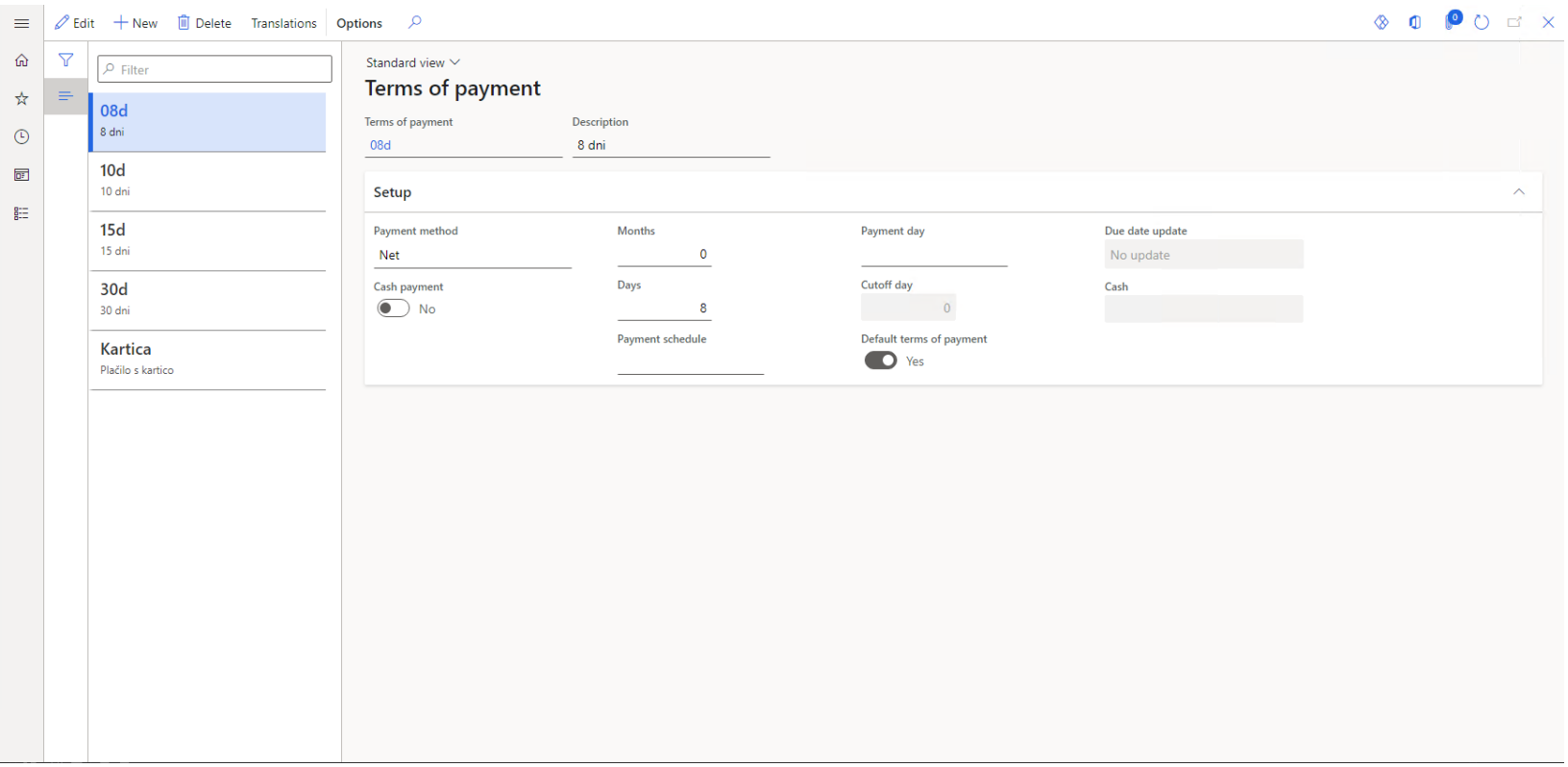
Define them by your own logic of how things should work.

## Form

| Name | Form Template | Label |
|---|---|---|
| AXMVehicleTypeParts | SimpleListDetails | Vehicle type parts |

**DataSource:** AXMVehicleTypeParts

The form should look like: AOT/Forms/PaymTerm




# Table

## Vehicle table -AXMVehicleTable

In this table we will create the vehicle.

### Properties

| Properties | |
|---|---|
| Name | AXMVehicleTable |
| FormRef | AXMVehicleTable |
| Label | Vehicles |
| Table Group | WorksheetHeader |
| Modified Date Time | Yes |
| Modified By | Yes |
| Created Date Time | Yes |
| Created By | Yes |

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## Fields

| Name | Field type | EDT | Properties |
|---|---|---|---|
| VehicleId | String | AXMVehicleId | Mandatory: Yes |
| VehicleColor | String | AXMVehicleColor | Mandatory: Yes |
| VehicleModel | String | | Mandatory: Yes |
| VehicleManufacurer | String | Name | Mandatory: Yes<br>This will be calculated field with lookup on the form. |
| VehicleType | Enum | AXMVehicleType | Mandatory: Yes |
| VehicleAssembleDate | Date | TransDate | Label: Assemble date |
| Status | Enum | AXMStatus | AllowEdit:No |

## Indexes

| Name | Field list | Properties |
|---|---|---|
| VehicleIdx | VehicleId | AllowDuplicates: No<br>AlternateKey: Yes |

**Additional requirements:**

Methods to be implemented: **find**() and **exists**().

Lookup method for VehicleManufacurer and the values will be taken from VendTable.

VehicleId needs to be number sequence in format: VEH-XXXXX.

**Field Groups:**

Define them by your own logic of how things should work.

# Table

## Vehicle lines - AXMVehicleLines

In this table we will insert the parts related to the vehicle on the header.

### Properties

| Properties | |
|---|---|
| Name | AXMVehicleLines |
| FormRef | AXMVehicleLines |
| Label | Vehicles |
| Table Group | WorksheetLines |
| Modified Date Time | Yes |
| Modified By | Yes |
| Created Date Time | Yes |
| Created By | Yes |

### Fields

| Name | Field type | EDT | Properties | Relation |
|---|---|---|---|---|
| LineNum | | LineNum | | |
| VehicleId | String | AXMVehicleId | Mandatory: Yes | AXMVehicleTable. AXMVehicleId |
| VehiclePartId | String | AXMVehiclePartID | Mandatory: Yes | AXMVehicleParts. VehiclePartId |
| VehiclePartName | String | AXMVehiclePart | Mandatory: Yes | |
| VehiclePartManufacurer | String | AXMVehicleManufacurer | Mandatory: Yes | |
| VehiclePartValidated | Enum | NoYesId | AllowEdit: No | |
| VehiclePartQty | Real | AXMPartQty | Mandatory: Yes Allow Negative: No | |
| VehiclePartPrice | Real | Price | | |

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### Indexes

| Name | Field list | Properties |
|---|---|---|
| VehicleLineIdx | VehicleId, VehiclePartId | AllowDuplicates: No<br>AlternateKey: Yes |

Methods to be implemented: **find**() and **exists**().

**Additional requirements:**

When you modify VehiclePartId it will automatically update VehiclePartName and VehiclePartPrice through event handler.

VehiclePartPrice will be calculated field based on quantity of the part in lines and the price of the part.

VehiclePartQty should not exceed vehicle part quantyti from VehicleParts table

**Field Groups:**

Define them by your own logic of how things should work.

# Form

## Vehicles list page form

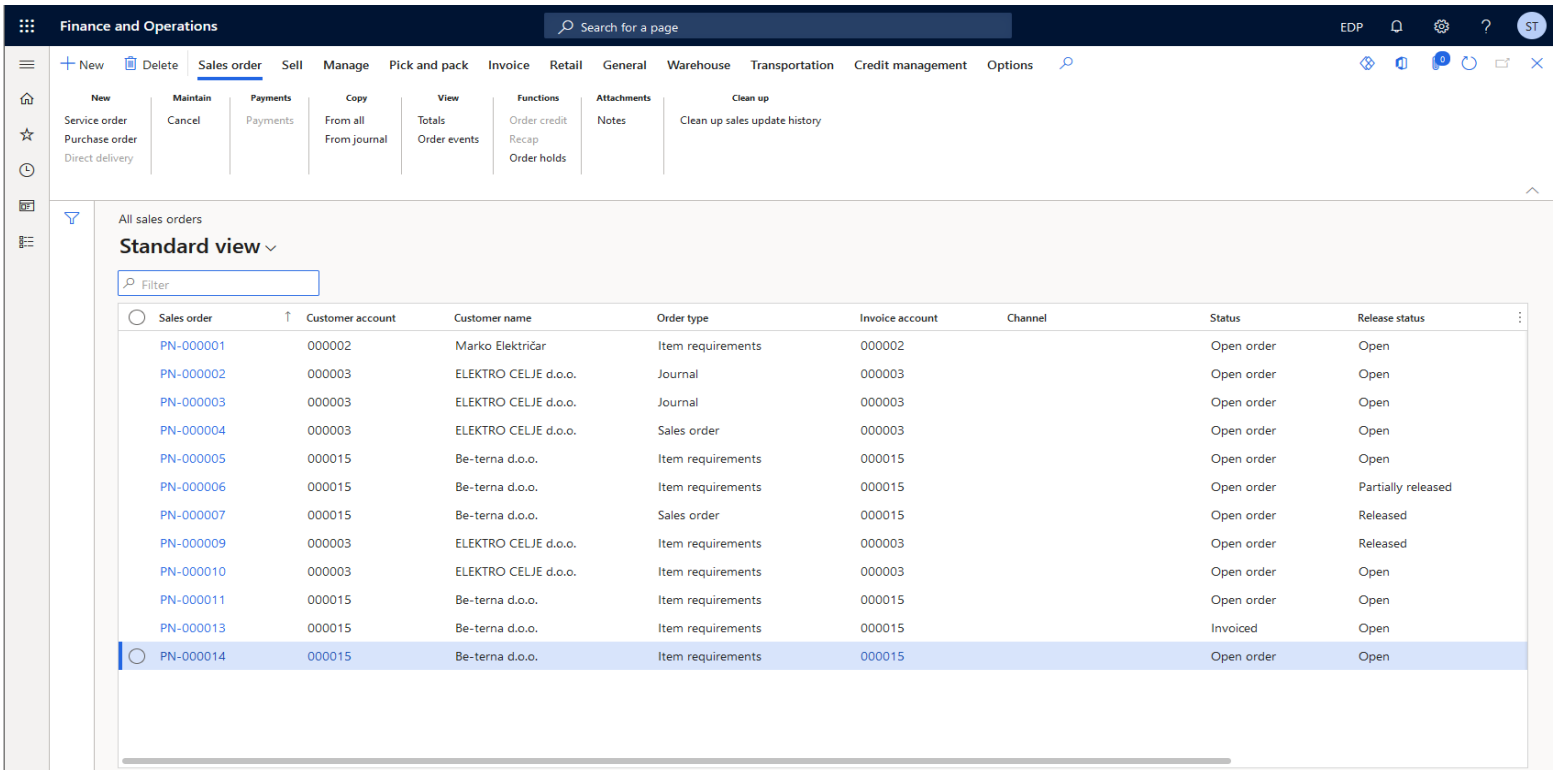
| Name | Form Template | Caption |
|---|---|---|
| AXMVehiclesListPage | List Page | Vehicles |

**Datasource**: For datasource we will use **AXMVehiclesListPage** query which is explained later in the document.

**Layout and functionalities**: The form should have a similar look as the image below. All of the fields from the header that will be shown on the form. The user should be able to add and delete records from this form. On the action pane add two buttons, one will be called “Validate Vehicle” and it will be menu item button that has link to the AXMValidateParts class. The second one will be called “Create Vehicle” and the menu item button will be assigned to the class AXMCreateVehicle – at first this button will be disable till all the lines are not validated.

Create a new New button that should look like SalesCreateOrder form (that is called when you click New on  SalesTableListPage) where it will open a new dialog and thought that will create a new record in Vehicle list page form.

**HINT**: Use form SalesTableListPage as a reference for creating this form




## Vehicles details form

| Name | Form Template | Label |
|---|---|---|
| AXMVehicles | DetailsTransaction | Vehicles |

**DataSource:** AXMVehicleTable, AXMVehicleLines

The form should look like: AOT/Forms/SalesTable

The details form will have the same action pane as the list page form.

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Edit | New | Delete | How to | Sales order | Sell | Manage | Pick and pack | Invoice | Retail | General | Warehouse

Sales order details | My view

# CSOR000003647 : Domestic Customer

Lines | Header | Open order

## Sales order header

**DELIVERY ADDRESS**

Name: Domestic Customer

Delivery address: Domestic Customer - Primary

Address:
primary street
1234 TT primary city
Netherlands

**BILLING ADDRESS**

Name

Billing Address: Domestic Customer - Primary

Address:
primary street
1234 TT primary city
Netherlands

**STATUTORY ADDRESS**

Name

Statutory address

Address

**DELIVERY DATE**

Requested ship date: 2/14/2022

Requested receipt date: 2/14/2022

Simulate delivery dates

Confirmed ship date

Confirmed receipt date

**REFERENCES**

Customer reference

Customer requisition

**DISCOUNTS**

Total discount %: 0.00

**WAREHOUSE**

Release status: Open

**TRANSPORTATION**

Routes

Edit | New | Delete | How to | Sales order | Sell | Manage | Pick and pack | Invoice | Retail | General | Warehouse

Sales order details | My view

CSOR000003647 : Domestic Customer

Lines | Header | Open order

Sales order header

## Sales order lines

Add line | Add lines | Add products | Remove | Sales order line | Financials | Inventory | Product and supply | Update line | Warehouse | Retail

| Type | Variant number | Product number | Item number | Product name | Sales category | CW quantit |
|---|---|---|---|---|---|---|
| | 9998-00001 | S000001000001 | S000001 | RSAT AFSACC 1202 65898554 / ... | | |
| | 9998-00001 | S000001000001 | S000001 | RSAT AFSACC 1202 65898554 / ... | | |

# Queries

### AXMVehiclesListPage

This query should be used in the list page form for listing vehicle by certain type. If the user clicks the menu item Vehicles by type/ Vehicles by status/Vehicles by date, A value should be added to this query such that vehicles will be filtered according to the selected value. This can be achieved by creating multiple queries and attaching them to menu items or via code, by dynamically adding a value to the original query in the method where query is initialized/executed (depending on which menu item the form was opened).

# Classes

### AXMValidateParts class

- Create a RunBaseBatch class that will check all of the lines from Vehicle lines for the given Vehicle and if the parts that are inserted satisfies for the vehicle to be created than turn all of the check boxes for the lines to true, show info message “All parts are successfully validated” and enable the “Create Vehicle” button on the Vehicle form. Also set the status on the header to Validated.

- For this class you need to create an action menu item and assign it to the button “Validate Vehicle” on the Vehicle form.

### AXMCreateVehicle class

- Create a class that will create an item in the Products form if all the lines are validated. Also, it will check if the assemble date on the vehicle is older than the date specified on the parameters form. If is older than the creating of the vehicle will fail and it will throw an error specifying that the cause of the error. If the vehicle is created, than set the status on the header to Completed.

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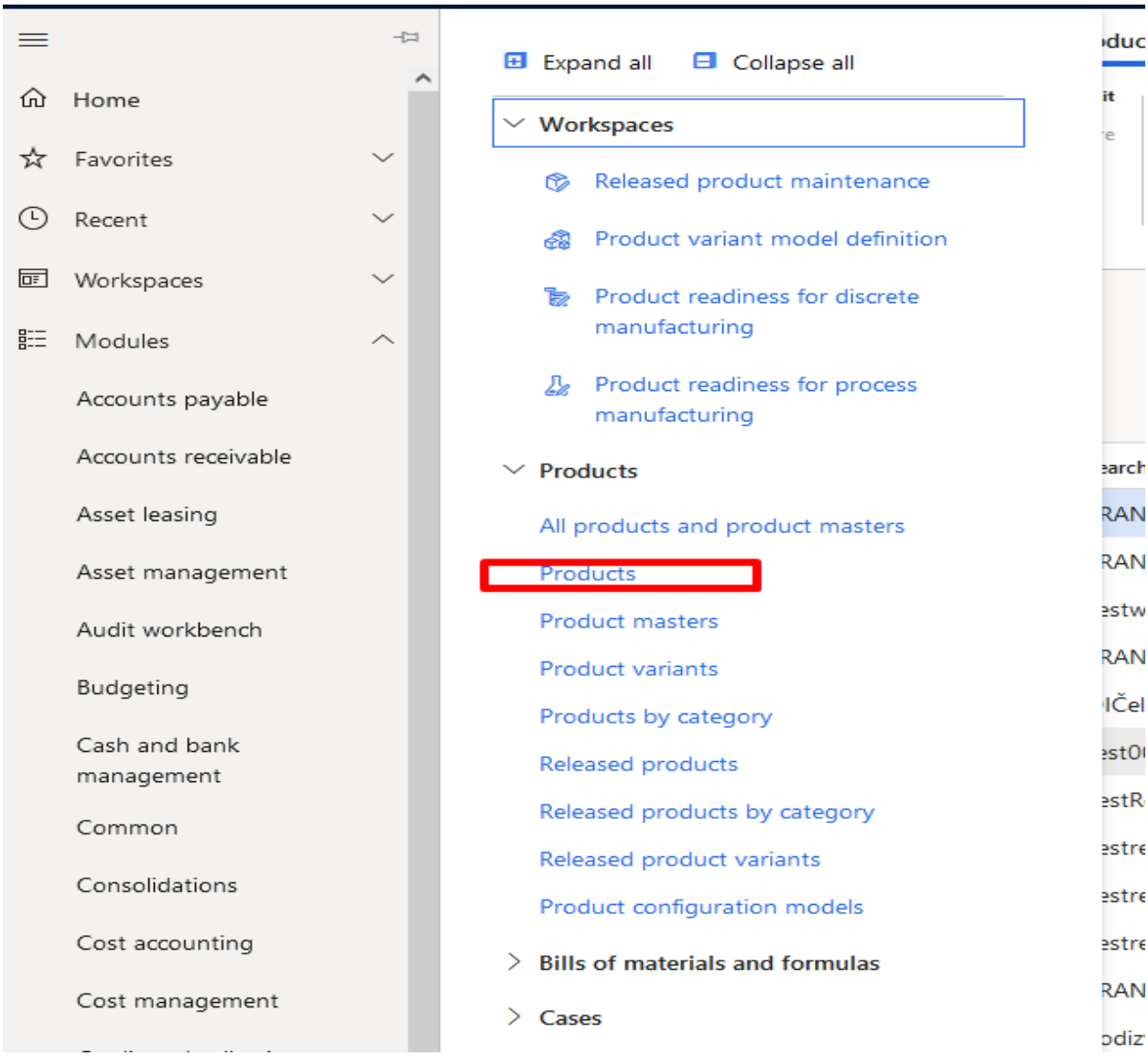


# D365 extensions

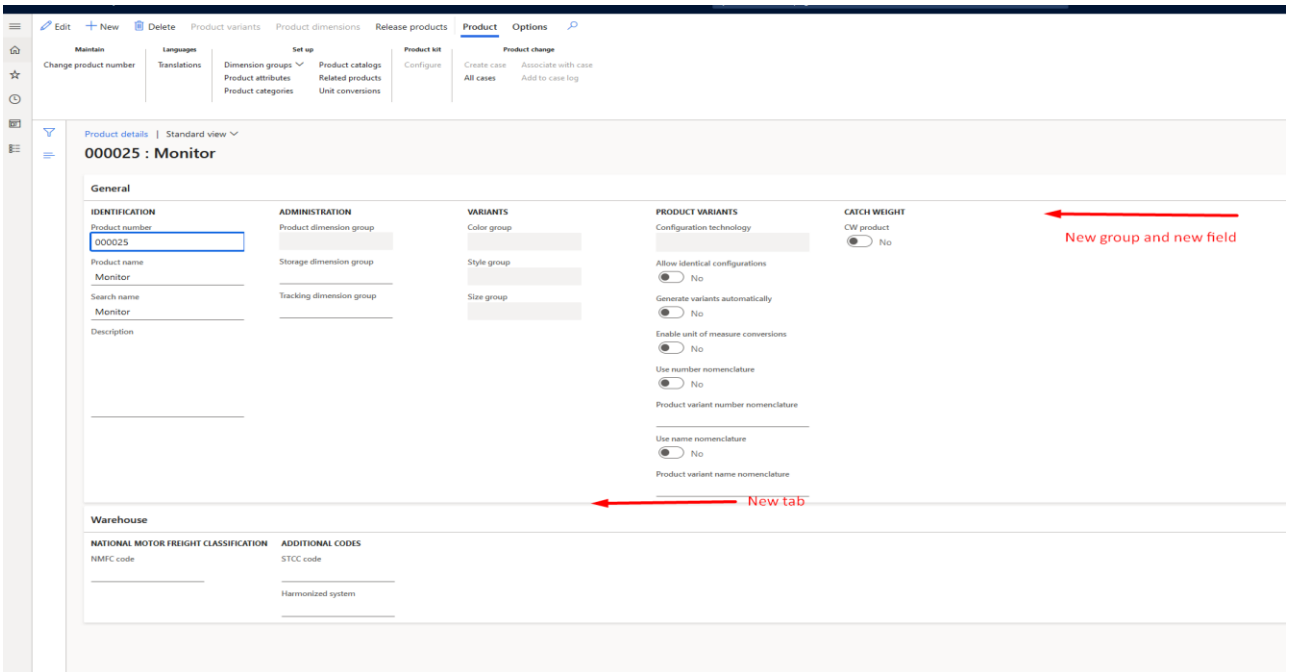
## EcoResProduct table

### Fields

| Name | Field type | EDT | Properties |
|---|---|---|---|
| VehiclePrice | Real | Price | Label: Vehicle price<br>AllowNegative: No |
| VehicleManufacturingDate | Date | TransDate | Label: Vehicle manufacturing date |
| IsItemVehicle | Enum | NoYesId | Label: Is item a vehicle?<br>AllowEdit: No |

## EcoResProductDetails form

On the EcoResProductDetails form extension, we need to create a new tab that will be located under the General tab, and it will have label Vehicle Manufacturing. Inside of the tab, add the newly created fields VehiclePrice and VehicleManufacturingDate. Also on in the General tab, add new group and then add the new IsItemVehicle checkbox.




## COC extension

Create a COC extension for active method on EcoResProduct datasouce from EcoResProductDetails form that will check if the IsItemVehicle is true or false. Based on that it will make the Vehicle Manufacturing tab visible or not.

# Reporting

**AXMManufacturedVehiclesReport report**

Create new SSRS report that will show manufactured vehicles based on certain period and type. The user should be able to choose from date, to date and type as a parameter to show the specific vehicles, also there should be a parameter field where user can enter threshold price amount. If the user leaves the parameters empty, the report will list all the vehicles. The report should have the following fields VehicleId, VehicleModel, VehicleType, VehiclePartName, VehiclePartQty, VehiclePartPrice.

Lines in the report should be grouped by VehicleId, VehicleModel, VehicleType.

If the VehiclePartPrice is greater than price entered on the parameter, the field should be with red background.

For the data for the report you will need to use the Report Data Provider class.

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# Menu items

| Name | Label | HelpText | ObjectType | Object |
|---|---|---|---|---|
| AXMVehicleManufacturerParameters | Vehicle manufacturer parameters | General parameters for Vehicle manufacturing. | Form | AXMVehicleManufacturerParameters |
| AXMVehicleParts | Vehicle parts | | Form | AXMVehicleParts |
| AXMVehicleTypeParts | Vehicle type parts | Parts that will be required for given type of vehicle to be build. | Form | AXMVehicleTypeParts |
| AXMVehiclesListPage | Vehicles | All vehicles | Form | AXMVehiclesListPage |
| AXMVehiclesByTypeListPage | List vehicles by type | | Form | AXMVehiclesListPage |
| AXMVehiclesByDateListPage | List vehicles by date | | Form | AXMVehiclesListPage |
| AXMVehiclesByStatusListPage | List vehicles by status | | Form | AXMVehiclesListPage |
| AXMValidateParts | Validate Vehicle | | Class | AXMValidateParts |
| AXMCreateVehicle | Create Vehicle | | Class | AXMCreateVehicle |
| AXMManufacturedVehiclesReport | Manufactured Vehicles | | SSRSReport | AXMManufacturedVehiclesReport |

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## Menu

Create new menu called: AXMVehicleManufacturer

Label: Vehicle manufacturer

The menu should have 4 areas:

Common: includes all list pages

Vehicle parts: Vehicle parts

Setup: includes Vehicle manufacturer parameters, Vehicle type parts

Reports: Manufactured Vehicles report

## Security

Create entry points for all menuitems